



COMP30026

Models of Computation

Cezary Kaliszyk and William Umboh

Lecture 2

Propositional Logic



Overview

Last time

- Overview of the subject
- A few problems solvable with logic
- Logical language

Today

- Formal propositional logic
- Soon: Mechanized proof

Propositional = Boolean Logic

Until the mid-19th century, “logic” meant Aristotelian logic.

George Boole took an **algebraic** view of logic.

Deep connection between logic and arithmetic.

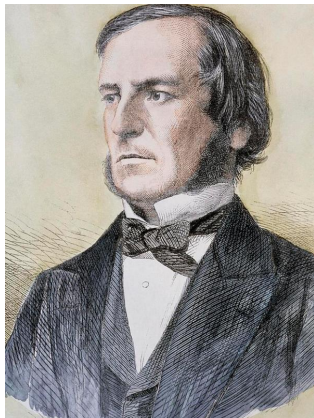


Figure: George Boole, circa 1864



Intro Puzzle

Heidi, Dina and Louise are being questioned by their aunt.

Here is what they say:

Heidi: "Dina and Louise had equal share in it; if one is guilty, so is the other."

Dina: "If Heidi is guilty, then so am I."

Louise: "Dina and I are not both guilty."

Their aunt, knowing that they are honest kids, realises that they cannot tell a lie.

Has she got sufficient information to decide who (if any) are guilty?



Definition

propositional **formulas** are built from

- **atoms** $P, Q, R, P_1, P_2, \dots$



Definition

propositional **formulas** are built from

- atoms $P, Q, R, P_1, P_2, \dots$
- **bottom** $\perp$ “false”



Definition

propositional **formulas** are built from

- | | | |
|--------------|----------------------------|---------|
| ■ atoms | $P, Q, R, P_1, P_2, \dots$ | |
| ■ bottom | $\perp$ | “false” |
| ■ top | $\top$ | “truth” |



Definition

propositional **formulas** are built from

- | | | |
|-------------------|----------------------------|---------------------|
| ■ atoms | $P, Q, R, P_1, P_2, \dots$ | |
| ■ bottom | $\perp$ | “false” |
| ■ top | $\top$ | “truth” |
| ■ negation | $\neg$ | $\neg P$ “not P ” |



Definition

propositional **formulas** are built from

■ atoms	$P, Q, R, P_1, P_2, \dots$	
■ bottom	$\perp$	“false”
■ top	$\top$	“truth”
■ negation	$\neg$	$\neg P$ “not P ”
■ conjunction	$\wedge$	$P \wedge Q$ “ P and Q ”

Definition

propositional **formulas** are built from

■ atoms	$P, Q, R, P_1, P_2, \dots$		
■ bottom	$\perp$		“false”
■ top	$\top$		“truth”
■ negation	$\neg$	$\neg P$	“not P ”
■ conjunction	$\wedge$	$P \wedge Q$	“ P and Q ”
■ disjunction	$\vee$	$P \vee Q$	“ P or Q ”



Definition

propositional **formulas** are built from

■ atoms	$P, Q, R, P_1, P_2, \dots$		
■ bottom	$\perp$		“false”
■ top	$\top$		“truth”
■ negation	$\neg$	$\neg P$	“not P ”
■ conjunction	$\wedge$	$P \wedge Q$	“ P and Q ”
■ disjunction	$\vee$	$P \vee Q$	“ P or Q ”
■ implication	$\rightarrow$	$P \rightarrow Q$	“if P then Q ”

Definition

propositional formulas are built from

■ atoms	$P, Q, R, P_1, P_2, \dots$	
■ bottom	$\perp$	“false”
■ top	$\top$	“truth”
■ negation	$\neg$	$\neg P$ “not P ”
■ conjunction	$\wedge$	$P \wedge Q$ “ P and Q ”
■ disjunction	$\vee$	$P \vee Q$ “ P or Q ”
■ implication	$\rightarrow$	$P \rightarrow Q$ “if P then Q ”

according to following:

$$\varphi ::= P \mid \perp \mid \top \mid (\neg \varphi) \mid (\varphi \wedge \varphi) \mid (\varphi \vee \varphi) \mid (\varphi \rightarrow \varphi)$$



Some Strictly Well-Formed Formulas

$$P \quad (1)$$

$$(P \rightarrow Q) \quad (2)$$

$$(P \vee (\neg P)) \quad (3)$$

$$(\neg(P \wedge (\neg P))) \quad (4)$$

$$(P \rightarrow (\neg P)) \quad (5)$$

$$(((P \rightarrow Q) \rightarrow P) \rightarrow P) \quad (6)$$



Some Strictly Well-Formed Formulas

$$P \quad (1)$$

$$(P \rightarrow Q) \quad (2)$$

$$(P \vee (\neg P)) \quad (3)$$

$$(\neg(P \wedge (\neg P))) \quad (4)$$

$$(P \rightarrow (\neg P)) \quad (5)$$

$$(((P \rightarrow Q) \rightarrow P) \rightarrow P) \quad (6)$$

Quiz

- Can we express an “if and only if” with what we have?
- $P \leftrightarrow Q$



Notational Conventions

Formulas can additionally follow these

- Omit outer parentheses



Notational Conventions

Formulas can additionally follow these

- Omit outer parentheses
- Negation binds stronger than $\wedge, \vee$



Notational Conventions

Formulas can additionally follow these

- Omit outer parentheses
- Negation binds stronger than $\wedge, \vee$
- $\wedge, \vee$ bind stronger than implication

Notational Conventions

Formulas can additionally follow these

- Omit outer parentheses
- Negation binds stronger than $\wedge, \vee$
- $\wedge, \vee$ bind stronger than implication
- $\rightarrow$ is right-associative:
 $P \rightarrow Q \rightarrow R$ denotes $P \rightarrow (Q \rightarrow R)$

Notational Conventions

Formulas can additionally follow these

- Omit outer parentheses
- Negation binds stronger than $\wedge, \vee$
- $\wedge, \vee$ bind stronger than implication
- $\rightarrow$ is right-associative:
 $P \rightarrow Q \rightarrow R$ denotes $P \rightarrow (Q \rightarrow R)$
- **Warning:** $P \wedge Q \vee R$ is AMBIGUOUS (so not well formed!)

We call these: **Well-Formed Formulas**

- e.g. $\neg P \wedge Q \rightarrow \neg R$



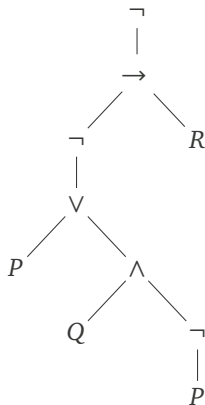
Examples (reminder: $\neg > \wedge, \vee > \rightarrow$)

formula $\neg(\neg(P \vee (Q \wedge \neg P)) \rightarrow R)$

parse tree

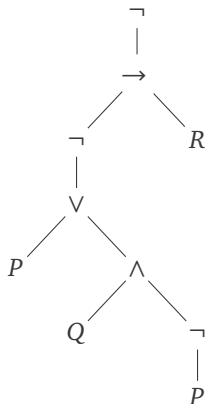
Examples (reminder: $\neg > \wedge, \vee > \rightarrow$)

formula $\neg(\neg(P \vee (Q \wedge \neg P)) \rightarrow R)$
parse tree



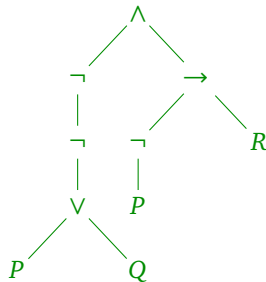
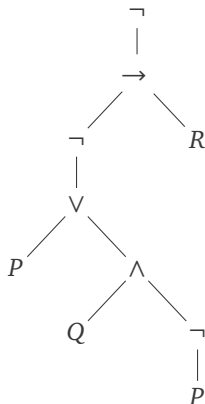
Examples (reminder: $\neg > \wedge, \vee > \rightarrow$)

formula $\neg(\neg(P \vee (Q \wedge \neg P)) \rightarrow R)$ $\neg\neg(P \vee Q) \wedge (\neg P \rightarrow R)$
 parse tree



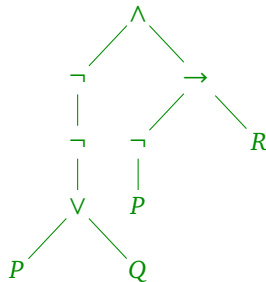
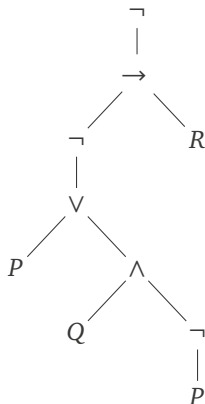
Examples (reminder: $\neg > \wedge, \vee > \rightarrow$)

formula $\neg(\neg(P \vee (Q \wedge \neg P)) \rightarrow R)$ $\neg\neg(P \vee Q) \wedge (\neg P \rightarrow R)$
 parse tree



Examples (reminder: $\neg > \wedge, \vee > \rightarrow$)

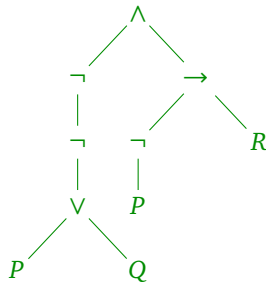
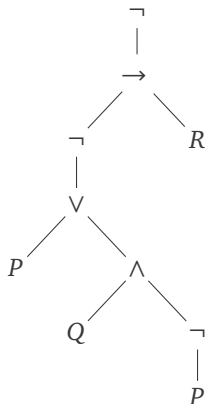
formula $\neg(\neg(P \vee (Q \wedge \neg P)) \rightarrow R)$ $\neg\neg(P \vee Q) \wedge (\neg P \rightarrow R)$
 parse tree



$$\neg\neg P \vee Q \wedge \neg P \rightarrow R$$

Examples (reminder: $\neg > \wedge, \vee > \rightarrow$)

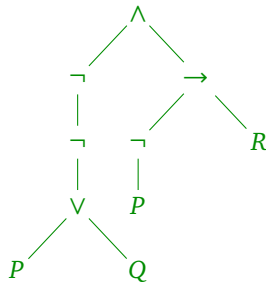
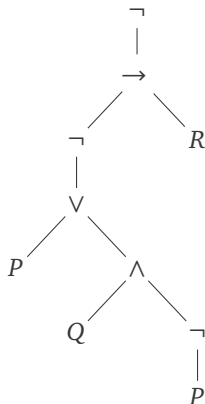
formula $\neg(\neg(P \vee (Q \wedge \neg P)) \rightarrow R)$ $\neg\neg(P \vee Q) \wedge (\neg P \rightarrow R)$
 parse tree



$$\neg\neg P \vee Q \wedge \neg P \rightarrow R$$

Examples (reminder: $\neg > \wedge, \vee > \rightarrow$)

formula $\neg(\neg(P \vee (Q \wedge \neg P)) \rightarrow R)$ $\neg\neg(P \vee Q) \wedge (\neg P \rightarrow R)$
 parse tree



$$\neg\neg P \vee Q \wedge \neg P \rightarrow R$$

$$\neg\neg(P \vee Q) \wedge \neg P \rightarrow R$$



Boolean Semantics: Connectives

Definition (Truth function)

A function from truth values to truth values.



Boolean Semantics: Connectives

Definition (Truth function)

A function from truth values to truth values.

Boolean truth values: **1** and **0**
(also written **t** and **f**, $\top$ and $\perp$).

Boolean Semantics: Connectives

Definition (Truth function)

A function from truth values to truth values.

Boolean truth values: **1** and **0**
(also written **t** and **f**, $\top$ and $\perp$).

Usually presented as a **truth table**:

A	B	$\neg A$	$A \wedge B$	$A \vee B$	$A \rightarrow B$	$A \leftrightarrow B$
0	0	1	0	0	1	1
0	1	1	0	1	1	0
1	0	0	0	1	0	0
1	1	0	1	1	1	1



Boolean Semantics: Letters

Propositional letters are **Boolean variables**.



Boolean Semantics: Letters

Propositional letters are **Boolean variables**.

Definition (Truth assignment, Valuation)

A function from propositional variables to truth values.



Boolean Semantics: Letters

Propositional letters are **Boolean variables**.

Definition (Truth assignment, Valuation)

A function from propositional variables to truth values.

Usual notation:

$$v = \{P \mapsto 1, Q \mapsto 0\}.$$



Boolean Semantics: Letters

Propositional letters are **Boolean variables**.

Definition (Truth assignment, Valuation)

A function from propositional variables to truth values.

Usual notation:

$$v = \{P \mapsto 1, Q \mapsto 0\}.$$

We then have:

$$v(P) = 1$$

$$v(Q) = v(R) = \dots = v(Z) = 0.$$



Truth of a Formula

Let $v = \{P \mapsto 1, Q \mapsto 0\}$.

Poll: Which of these formulas are true under v ?

1. $P \wedge Q$
2. $(P \vee Q) \wedge (P \vee R)$
3. $P \rightarrow Q$
4. $\neg P \rightarrow \neg Q$



Truth of a Formula

Let $v = \{P \mapsto 1, Q \mapsto 0\}$.

Poll: Which of these formulas are true under v ?

1. $P \wedge Q$
2. $(P \vee Q) \wedge (P \vee R)$
3. $P \rightarrow Q$
4. $\neg P \rightarrow \neg Q$

Shorthand: “ $v \models \phi$ ” means “ ϕ is true under v ”.

Truth Tables for Formulas

P	Q	R	$((P \wedge Q) \vee R)$				
0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	1
0	1	0	0	0	1	0	0
0	1	1	0	0	1	1	1
1	0	0	1	0	0	0	0
1	0	1	1	0	0	1	1
1	1	0	1	1	1	1	0
1	1	1	1	1	1	1	1



Poll

Which of these have the same truth tables?

1. $P \rightarrow Q$ and $\neg Q \rightarrow \neg P$
2. $(P \rightarrow Q) \wedge (P \rightarrow R)$ and $P \rightarrow (Q \wedge R)$
3. $(P \rightarrow R) \wedge (Q \rightarrow R)$ and $(P \wedge Q) \rightarrow R$

Hint: $P \rightarrow Q$ has the same truth table as $\neg P \vee Q$.



Logical Equivalence

Definition

Formulas are *logically equivalent* iff they have equal truth values under **every** truth assignment.

Shorthand: " $F \equiv G$ " means " F is logically equivalent to G ".



Warning: “ $\rightarrow$ ” is weird!

Often read as “implies”, but causality is not required!

A	B	$A \rightarrow B$
0	0	1
0	1	1
1	0	0
1	1	1

1. If there are no bugs, then the program runs correctly.



Warning: “ $\rightarrow$ ” is weird!

Often read as “implies”, but causality is not required!

A	B	$A \rightarrow B$
0	0	1
0	1	1
1	0	0
1	1	1

1. If there are no bugs, then the program runs correctly.
2. If Melbourne is in Queensland, then Brisbane is in Victoria.



Modus Ponens

$$\frac{P \rightarrow Q \quad P}{Q}$$

A rule is **sound** if every model of the premises is a model of the conclusion.

Challenge: prove that modus ponens is sound.